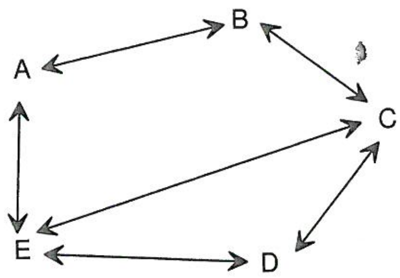
26. The diagram below shows the road routes between the five towns A, B, C, D and E.



(a) Represent the the above information in a stage one matrix M.

(b) Use a matrix method to find the number of routes between any two towns via another town.

(c) Use a matrix method to find the number of routes between any two towns via two other towns.

(d) Determine the matrix $M + M^2$.

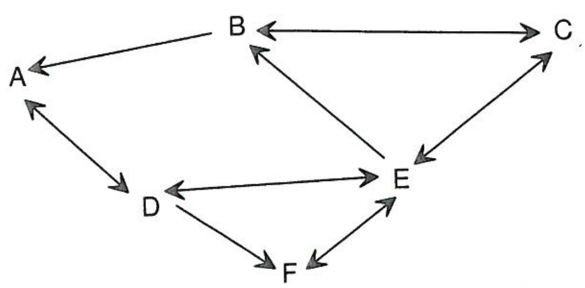
(e) What information is given by the matrix $M + M^2$?

(f) Determine the matrix $M + M^2 + M^3$.

(g) What information is given by the matrix $M + M^2 + M^3$?

(h) A new road route is constructed linking A and C. What effect will this have if it is decided to choose one town as a service centre for these five towns?

27. Consider six cities linked by rail routes as shown.

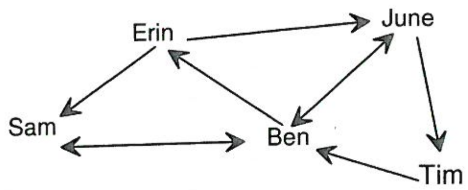


(a) Construct a matrix that represents all of the direct routes between these cities. Name this matrix R.

(b) Construct the matrix that represents all the routes with one stop over.

(c) Construct the matrix that represents all routes with at most one stop over.

3. A team of five students working on a science project can only get in contact with some of the group outside of school hours.
The diagram on the right shows that Erin can directly only contact June and Sam, whereas Ben can directly directly contact Erin, June and Sam, etc.



(a) Is the given network a directed or undirected network? Explain your choice.

(b) Construct a matrix which shows the direct contact(one stage matrix) between the members of this team.

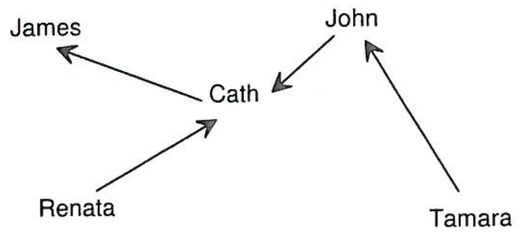
(c) Construct a two stage matrix showing indirect contact between the members of this team.

(d) Square your stage one matrix.

(e) Compare your two stage matrix with the square of your stage one matrix. Discuss

4. The network shows the relationship "is a parent of". For example, Cath is a parent of James, Renata is a parent of Cath etc.

(a) Construct a matrix P which models the information contained in the network.



(b) Construct matrix P^2. What information is contained in P^2?

(c) Construct matrix P^3. What information is contained in P^3?

26. (a) $$M = \begin{array}{cc} & \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

(b) $$M^2 = \begin{array}{cc} & \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} & \begin{bmatrix} 2 & 0 & 2 & 1 & 0 \\ 0 & 2 & 0 & 1 & 2 \\ 2 & 0 & 3 & 1 & 1 \\ 1 & 1 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 & 3 \end{bmatrix} \end{array}$$

(c) $$M^3 = \begin{array}{cc} & \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} & \begin{bmatrix} 0 & 4 & 1 & 2 & 5 \\ 4 & 0 & 5 & 2 & 1 \\ 1 & 5 & 2 & 4 & 6 \\ 2 & 2 & 4 & 2 & 4 \\ 5 & 1 & 6 & 4 & 2 \end{bmatrix} \end{array}$$

(d) $$M+M^2 = \begin{array}{cc} & \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} & \begin{bmatrix} 2 & 1 & 2 & 1 & 1 \\ 1 & 2 & 1 & 1 & 2 \\ 2 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 2 & 2 & 3 \end{bmatrix} \end{array}$$

(e) This matrix shows the number of routes with at most one stop over, that is at most one town between any two given towns.

(f) $$M + M^2 + M^3 = \begin{array}{cc} & \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} & \begin{bmatrix} 2 & 5 & 3 & 3 & 6 \\ 5 & 2 & 6 & 3 & 3 \\ 3 & 6 & 5 & 6 & 8 \\ 3 & 3 & 6 & 4 & 6 \\ 6 & 3 & 8 & 6 & 5 \end{bmatrix} \end{array}$$

(g) This matrix shows the number of routes with at most two stop overs, that is at most two towns between any two given towns.

(h) Town will be chosen as a service centre as it directly links all towns.

27. (a) $$\begin{array}{cc} & \begin{array}{cccccc} A & B & C & D & E & F \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \\ F \end{array} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

(b) $$\begin{array}{cc} & \begin{array}{cccccc} A & B & C & D & E & F \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \\ F \end{array} & \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 0 & 3 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

(c) $$\begin{array}{cc} & \begin{array}{cccccc} A & B & C & D & E & F \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \\ F \end{array} & \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 2 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 2 & 2 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

3. (a) Directed network because the number of direct routes from any vertex X to vertex Y is not the same as the number of direct routes from vertex Y to vertex X. In other words there are one way paths in the network.

(b). $$\begin{array}{cc} & \begin{array}{ccccc} \text{Ben} & \text{Erin} & \text{June} & \text{Sam} & \text{Tim} \end{array} \\ \begin{array}{c} \text{Ben} \\ \text{Erin} \\ \text{June} \\ \text{Sam} \\ \text{Tim} \end{array} & \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

(c) $$\begin{array}{cc} & \begin{array}{ccccc} \text{Ben} & \text{Erin} & \text{June} & \text{Sam} & \text{Tim} \end{array} \\ \begin{array}{c} \text{Ben} \\ \text{Erin} \\ \text{June} \\ \text{Sam} \\ \text{Tim} \end{array} & \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 2 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

(d) $$\begin{array}{cc} & \begin{array}{ccccc} \text{Ben} & \text{Erin} & \text{June} & \text{Sam} & \text{Tim} \end{array} \\ \begin{array}{c} \text{Ben} \\ \text{Erin} \\ \text{June} \\ \text{Sam} \\ \text{Tim} \end{array} & \begin{bmatrix} 2 & 0 & 1 & 1 & 1 \\ 2 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

(e) Only the entries in the leading diagonal differ. The two stage matrix has only zeros in the leading diagonal which is not the case for the stage one matrix squared.

4. (a) $$P = \begin{array}{cc} & \begin{array}{ccccc} \text{Cath} & \text{James} & \text{John} & \text{Renata} & \text{Tamara} \end{array} \\ \begin{array}{c} \text{Cath} \\ \text{James} \\ \text{John} \\ \text{Renata} \\ \text{Tamara} \end{array} & \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

(b) $$P^2 = \begin{array}{cc} & \begin{array}{ccccc} \text{Cath} & \text{James} & \text{John} & \text{Renata} & \text{Tamara} \end{array} \\ \begin{array}{c} \text{Cath} \\ \text{James} \\ \text{John} \\ \text{Renata} \\ \text{Tamara} \end{array} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Matrix P^2 shows that John and Renata are grandparents of James and Tamara is a grandparent of Cath.

(c) $$P^3 = \begin{array}{cc} & \begin{array}{ccccc} \text{Cath} & \text{James} & \text{John} & \text{Renata} & \text{Tamara} \end{array} \\ \begin{array}{c} \text{Cath} \\ \text{James} \\ \text{John} \\ \text{Renata} \\ \text{Tamara} \end{array} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Matrix P^3 shows that Tamara is a great grandparent of James.